Never Republish Again!

Using the ArcGIS API for Python to Modify and Manage Hosted Feature Layers

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HI, I'M YOUR HOST!

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GIS Intern - 3 Years
MS in Geography - SIUE
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Why do I hate republishing?

For static data layers republishing is not that bad. However:

- Time consuming
- Have to drop and recreate views
- Adjust dependencies and share accordingly after that

For editable layers:

- Edits must be pulled down before republishing
- GlobalIDs must be maintained for relationships
- Even more time consuming
- Extra steps to do things that should be simple additions

Biggest reason for doing so? Not out of the nature of republishing itself, but because ArcGIS Online often does not provide the option to.

Enter: The ArcGIS API for Python

ArcPy vs ArcGIS API for Python

ArcPy

- Essentially ArcGIS Pro but tools are called through code
- Requires a licensed environment (ArcGIS Pro/ArcGIS Server/Enterprise)
- Working with geodatabases, layouts, files on network
- Generally preferable for spatial analysis
- Cursors

ArcGIS API for Python

- An API (application programming interface) for ArcGIS Online/Enterprise
- Does not require a licensed environment (standalone servers, locally, notebooks, etc.)
- Working with web hosted content, users, groups, etc
- Light, quick analysis
- Data frames

Why use the API for Python for AGO Content management?

- Provides access to capabilities not immediately available in AGO
- Programmatic management of data standards
- Automate repetitive tasks
- Take advantage of open source libraries and capabilities

In the last two years, I have used the API to do the following:

- Generated over 2000 field descriptions using AI
- Created a web application for managing credits, users, groups, for all of our client organizations
- Deploy application stacks and updates at the push of a button
- Detect outages from the RSS feed and update Hubs with error messages that services may be unavailable
- Built a scalable backup utility for our clients when software costs rose

ArcGIS Online slogs updates with clicks, load screens, and hiding capabilities.

My goal: get you thinking on how the API can help your workflows, without you republishing every time

Let's Get Started!

We will walk through 5 scenarios, coding together

Notebooks are available in the following GitHub repository if you fall behind:

https://tinyurl.com/5e5thrp6

Scenario 1

```
Hello NERD!

I am the forester for your client, and I need a web map
of my trees STAT.

I also need a public layer of my trees for them to view.

Hurry up,
The CLIENT
```

Takeaways from Scenario 1

Use gis("home") to connect to ArcGIS Online from a notebook

Instead of "home", may use password/username, profiles, or other options depending on the environment

Always import the ArcGIS API for Python using from arcgis.gis import GIS

Hosted Feature Layers have three places configuration is stored: Properties, Data, and the Service Definition

Create folders using gis.content.folders.create()

gis.content.clone_items() will copy an item and all dependencies (use with caution)

ArcGIS Online items are accessed by ItemID and gis.content.get()

Use the ArcGIS Online Assistant, it is an extremely helpful tool.

Scenario 2

```
Hello again NERD!

Thanks for getting me those trees, no idea how you did
that so fast. Problem is, you done messed up.

My trees need to have more than just the 30 tree names
for the drop down list. I need another 20 names
included.

The list is attached.

Hurry up,
The CLIENT
```

Takeaways from Scenario 2

The pandas package and dataframes are immensely helpful for GIS (geopandas anyone?)

To interact with an individual hosted feature layer, use FeatureLayer() and a URL

Use layer.properties to get the service definition from a FeatureLayer() object

The attribute information are stored in a property called “fields”

We can loop through the array of fields and check the “name” property to ensure we are updating the one we want.

Use layer.manager.update_definition() to update the field

Changes are made in memory, than pushed to the HFL

Printing the definitions will allow us to better understand what we are updating.

Scenario 3

View 1
”sourceSchemaChangesAllowed”: True

Joined View 1
”sourceSchemaChangesAllowed”: True

Hosted Feature Layer
”sourceSchemaChangesAllowed”: False

View 2
”sourceSchemaChangesAllowed”: False

Takeaways from Scenario 3

If we want to get all the layers for a hosted feature service, we use FeatureLayerCollection() and ItemID

We can search for items in AGO using gis.content.search() (use with caution)

We can query for views using the query 'type:"Feature Service" AND typekeywords:"View Service"'

The hosted feature services for a view is stored in the "related_items" property of a view

We can use try/except blocks to attempt to do things in a loop without stopping the whole process

To update the definition of a hosted feature service, we still use manager.update_definition()

"sourceSchemaChangesAllowed": True allows for changing a feature service data schema, and is based on all the views of a service

Changing this property does mean any changes are not reflected in the views

Scenario 4

```
Hello associate,

Thanks for the updates for that drop down, me and the
boys are filling in the trees real nice.

I was wondering if it was possible to use your fancy
schmancy ArcGIS Online to record inspections on the
trees as well? We just need to note ash borer signs
and general condition of the tree.

Thanks,
The Client
```

Takeaways from Scenario 4

Use other libraries like "copy" to save on writing your own code

The easiest way to add a table to an existing hosted feature service is to make one and copy it over

Keep track of if you need to use the index of an array or the HFL ID, the "first table" [0] could have an id of 2 on the REST endpoint

Convert properties to a dictionary using dict()

When adding a new object instead of changing an existing property, we need to use manager.add_to_definition()

When adding a table, we need to update both the definition and the data property in AGO to see the changes

Use .update() to update the properties of and AGOL item (in contrast to a feature service definition)

Table MUST be published from ArcGIS Pro, a table in AGO is missing the data property (bug?)

Scenario 5

Creating the

RELATIONSHIP

Takeaways from Scenario 5

Layers and tables are stored differently on the Feature Layer Collection

Need to create two relationships, one for each “side”

The definitions use the ID from the hosted feature service

“Name” refers to how the relationship appears in AGO, not the name of the tables

Reference another hosted feature service with the desired relationship type to make it easier

Only included necessary properties when building the relationship to avoid it getting too complicated

Tips and Tricks to the API for Python

- Avoid using LLMs/AI to generate code to start
  - ESRI is constantly changing the API, and as a result the current models often use deprecated or nonexistent tools
  - Use for other Python functions or capabilities
  - If you are skilled enough to recognize these flaws, than it can be useful
  - They WILL get better
- Use the documentation
  - Not as good as the documentation on ArcPy
  - The emphasis on classes and methods vs functions can be difficult for GIS staff like us
- Working with complicated items can be difficult (Dashboards, Hubs, Experience Builders)
  - Take the time to read through the JSON objects and understand how the stack relates to each other
- Use the ArcGIS Online Assistant (new version) and Admin tools
- Use Google to research as much as possible, that's what programmers do :)
- Jupyter Notebooks are nice, but I do recommend an IDE/other environment

Bonus Round

Hello GIS expert,

Thank you so much for the new tools, things are working much faster now due to your efforts.

I have one last request. I want to add the value of "Zelkova" to the drop down list. Also, if you could sort the list, it would be greatly appreciated.

Thanks,
Jimmy

Any

QUESTIONS?

That’s all folks!

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THANK YOU!